

UPSC Civil Services Preliminary Examination - CSAT Paper II

DETAILED SOLUTIONS / ANSWER KEY | Time: 2 Hours | Maximum Marks: 200

Quick Key

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Detailed solutions

Q1. Answer: A | Reading Comprehension

Passage: A hospital introduced online appointments. Missed visits fell for smartphone users but rose among patients who shared phones. A call-based fallback restored access without abandoning the digital system. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

- A. Digital services need fallback channels to avoid exclusion. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.
- B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.
- C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.
- D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q2. Answer: B | Reading Comprehension

Passage: A hospital introduced online appointments. Missed visits fell for smartphone users but rose among patients who shared phones. A call-based fallback restored access without abandoning the digital system. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

- A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.
- B. A uniform channel can distribute convenience unequally. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.
- C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q3. Answer: C | Reading Comprehension

Passage: A hospital introduced online appointments. Missed visits fell for smartphone users but rose among patients who shared phones. A call-based fallback restored access without abandoning the digital system. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Shared-phone users could use the call-based alternative. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q4. Answer: D | Reading Comprehension

Passage: A hospital introduced online appointments. Missed visits fell for smartphone users but rose among patients who shared phones. A call-based fallback restored access without abandoning the digital system. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. pragmatic - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q5. Answer: A | Reading Comprehension

Passage: A watershed project measured only the number of check dams. An audit found that poorly located structures trapped little runoff, while fewer structures placed after contour mapping improved recharge. Counting assets obscured performance. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Asset counts are weaker than outcome and location-sensitive evaluation. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q6. Answer: B | Reading Comprehension

Passage: A watershed project measured only the number of check dams. An audit found that poorly located structures trapped little runoff, while fewer structures placed after contour mapping improved recharge. Counting assets obscured performance. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. More structures do not necessarily produce more recharge. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q7. Answer: C | Reading Comprehension

Passage: A watershed project measured only the number of check dams. An audit found that poorly located structures trapped little runoff, while fewer structures placed after contour mapping improved recharge. Counting assets obscured performance. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Contour-informed placement affects captured runoff. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q8. Answer: D | Reading Comprehension

Passage: A watershed project measured only the number of check dams. An audit found that poorly located structures trapped little runoff, while fewer structures placed after contour mapping improved recharge. Counting assets obscured performance. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. critical but constructive - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q9. Answer: A | Reading Comprehension

Passage: A public library extended opening hours. Usage increased mainly after bus timings were coordinated with closing time. The library had removed a formal barrier, while transport coordination removed a practical one. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Access depends on complementary services, not formal availability alone. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q10. Answer: B | Reading Comprehension

Passage: A public library extended opening hours. Usage increased mainly after bus timings were coordinated with closing time. The library had removed a formal barrier, while transport coordination removed a practical one. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Longer hours may be ineffective when return transport is unavailable. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q11. Answer: C | Reading Comprehension

Passage: A public library extended opening hours. Usage increased mainly after bus timings were coordinated with closing time. The library had removed a formal barrier, while transport coordination removed a practical one. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Some potential users depended on buses. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q12. Answer: D | Reading Comprehension

Passage: A public library extended opening hours. Usage increased mainly after bus timings were coordinated with closing time. The library had removed a formal barrier, while transport coordination removed a practical one. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. explanatory - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q13. Answer: A | Reading Comprehension

Passage: Farmers received weather advisories by text. Adoption improved when messages specified the crop stage and local rainfall probability rather than giving district-wide warnings. More information was not enough; relevant information changed decisions. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Advisories influence action when they are locally and temporally relevant. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q14. Answer: B | Reading Comprehension

Passage: Farmers received weather advisories by text. Adoption improved when messages specified the crop stage and local rainfall probability rather than giving district-wide warnings. More information was not enough; relevant information changed decisions. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Information quantity and decision usefulness are not identical. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q15. Answer: C | Reading Comprehension

Passage: Farmers received weather advisories by text. Adoption improved when messages specified the crop stage and local rainfall probability rather than giving district-wide warnings. More information was not enough; relevant information changed decisions. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Farmers could relate crop-stage advice to an available action. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q16. Answer: D | Reading Comprehension

Passage: Farmers received weather advisories by text. Adoption improved when messages specified the crop stage and local rainfall probability rather than giving district-wide warnings. More information was not enough; relevant information changed decisions. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. diagnostic - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q17. Answer: A | Reading Comprehension

Passage: A municipal dashboard published complaint totals. Departments first closed easy cases to improve rankings. After the metric included age and recurrence of complaints, effort shifted towards persistent failures. Measurement changed behaviour twice. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Performance metrics must anticipate strategic responses. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q18. Answer: B | Reading Comprehension

Passage: A municipal dashboard published complaint totals. Departments first closed easy cases to improve rankings. After the metric included age and recurrence of complaints, effort shifted towards persistent failures. Measurement changed behaviour twice. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Simple closure counts can reward superficial compliance. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q19. Answer: C | Reading Comprehension

Passage: A municipal dashboard published complaint totals. Departments first closed easy cases to improve rankings. After the metric included age and recurrence of complaints, effort shifted towards persistent failures. Measurement changed behaviour twice. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Departments respond to the incentives embedded in rankings. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q20. Answer: D | Reading Comprehension

Passage: A municipal dashboard published complaint totals. Departments first closed easy cases to improve rankings. After the metric included age and recurrence of complaints, effort shifted towards persistent failures. Measurement

changed behaviour twice. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. analytical - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q21. Answer: A | Numeracy

What is the remainder when $171^2 + 3 \times 171 + 5$ is divided by 9?

A. 5 - CORRECT: Correct. Reduce 171 modulo 9, then evaluate $r^2 + 3r + 5$ modulo 9; the remainder is 5.

B. 6 - INCORRECT: Incorrect: 6 does not follow the stated data. Reduce 171 modulo 9, then evaluate $r^2 + 3r + 5$ modulo 9; the remainder is 5.

C. 7 - INCORRECT: Incorrect: 7 is an intermediate, sign or operation error. Reduce 171 modulo 9, then evaluate $r^2 + 3r + 5$ modulo 9; the remainder is 5.

D. 8 - INCORRECT: Incorrect: 8 ignores a condition or uses the wrong base. Reduce 171 modulo 9, then evaluate $r^2 + 3r + 5$ modulo 9; the remainder is 5.

Method / explanation: Reduce 171 modulo 9, then evaluate $r^2 + 3r + 5$ modulo 9; the remainder is 5.

Trap: Reduce each term modulo the divisor; do not expand the large square unnecessarily.

Q22. Answer: B | Numeracy

Two numbers are obtained by doubling 22 and tripling 36. What is their HCF?

A. 5 - INCORRECT: Incorrect: 5 does not follow the stated data. The derived numbers are 44 and 108; Euclid's algorithm gives $\gcd(44, 108) = 4$.

B. 4 - CORRECT: Correct. The derived numbers are 44 and 108; Euclid's algorithm gives $\gcd(44, 108) = 4$.

C. 6 - INCORRECT: Incorrect: 6 is an intermediate, sign or operation error. The derived numbers are 44 and 108; Euclid's algorithm gives $\gcd(44, 108) = 4$.

D. 1188 - INCORRECT: Incorrect: 1188 ignores a condition or uses the wrong base. The derived numbers are 44 and 108; Euclid's algorithm gives $\gcd(44, 108) = 4$.

Method / explanation: The derived numbers are 44 and 108; Euclid's algorithm gives $\gcd(44, 108) = 4$.

Trap: Apply the transformation before computing the HCF.

Q23. Answer: C | Numeracy

How many positive divisors does 72 have?

A. 11 - INCORRECT: Incorrect: 11 does not follow the stated data. $72 = 2^3 \times 3^2$, so divisor count = $(4)(3) = 12$.

B. 13 - INCORRECT: Incorrect: 13 is an intermediate, sign or operation error. $72 = 2^3 \times 3^2$, so divisor count = $(4)(3) = 12$.

C. 12 - CORRECT: Correct. $72 = 2^3 \times 3^2$, so divisor count = $(4)(3) = 12$.

D. 7 - INCORRECT: Incorrect: 7 ignores a condition or uses the wrong base. $72 = 2^3 \times 3^2$, so divisor count = $(4)(3) = 12$.

Method / explanation: $72 = 2^3 \times 3^2$, so divisor count = $(4)(3) = 12$.

Trap: Add one to every prime exponent before multiplying.

Q24. Answer: D | Numeracy

What is the unit digit of $(5^{13}) \times (9^7)$?

A. 6 - INCORRECT: Incorrect: 6 does not follow the stated data. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 5.

B. 7 - INCORRECT: Incorrect: 7 is an intermediate, sign or operation error. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 5.

C. 8 - INCORRECT: Incorrect: 8 ignores a condition or uses the wrong base. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 5.

D. 5 - CORRECT: Correct. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 5.

Method / explanation: Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 5.

Trap: Do not add the unit digits of the factors.

Q25. Answer: A | Numeracy

The arithmetic sequence begins 5, 9, 13. What is the sum of its first 9 terms?

A. 189.0 - CORRECT: Correct. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 5 + 8 \times 4] = 189.0$.

B. 185.0 - INCORRECT: Incorrect: 185.0 does not follow the stated data. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 5 + 8 \times 4] = 189.0$.

C. 193.0 - INCORRECT: Incorrect: 193.0 is an intermediate, sign or operation error. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 5 + 8 \times 4] = 189.0$.

D. 37 - INCORRECT: Incorrect: 37 ignores a condition or uses the wrong base. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 5 + 8 \times 4] = 189.0$.

Method / explanation: $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 5 + 8 \times 4] = 189.0$.

Trap: The last term is not the sum.

Q26. Answer: B | Numeracy

After multiplying 10098 by 9, what is the sum of the decimal digits of the product?

A. 26 - INCORRECT: Incorrect: 26 does not follow the stated data. $10098 \times 9 = 90882$; adding the digits of 90882 gives 27.

B. 27 - CORRECT: Correct. $10098 \times 9 = 90882$; adding the digits of 90882 gives 27.

C. 28 - INCORRECT: Incorrect: 28 is an intermediate, sign or operation error. $10098 \times 9 = 90882$; adding the digits of 90882 gives 27.

D. 29 - INCORRECT: Incorrect: 29 ignores a condition or uses the wrong base. $10098 \times 9 = 90882$; adding the digits of 90882 gives 27.

Method / explanation: $10098 \times 9 = 90882$; adding the digits of 90882 gives 27.

Trap: The divisibility-by-9 check verifies the digit sum but does not by itself give its unreduced value.

Q27. Answer: C | Numeracy

How many integers from 17 through 124, inclusive, are divisible by 8 but not by 16?

A. 6 - INCORRECT: Incorrect: 6 does not follow the stated data. Count multiples of 8 (13) and subtract multiples of 16 (6); result 7.

B. 8 - INCORRECT: Incorrect: 8 is an intermediate, sign or operation error. Count multiples of 8 (13) and subtract multiples of 16 (6); result 7.

C. 7 - CORRECT: Correct. Count multiples of 8 (13) and subtract multiples of 16 (6); result 7.

D. 9 - INCORRECT: Incorrect: 9 ignores a condition or uses the wrong base. Count multiples of 8 (13) and subtract multiples of 16 (6); result 7.

Method / explanation: Count multiples of 8 (13) and subtract multiples of 16 (6); result 7.

Trap: Multiples of the larger divisor are a subset and must be excluded once.

Q28. Answer: D | Numeracy

The values are 14, 20, 26, 32. If the smallest falls by 2 and the largest rises by 4, what is the new mean?

A. 23.0 - INCORRECT: Incorrect: 23.0 does not follow the stated data. The adjusted values are [12, 20, 26, 36]; their total 94 divided by 4 gives 23.5.

B. 24.5 - INCORRECT: Incorrect: 24.5 is an intermediate, sign or operation error. The adjusted values are [12, 20, 26, 36]; their total 94 divided by 4 gives 23.5.

C. 25.5 - INCORRECT: Incorrect: 25.5 ignores a condition or uses the wrong base. The adjusted values are [12, 20, 26, 36]; their total 94 divided by 4 gives 23.5.

D. 23.5 - CORRECT: Correct. The adjusted values are [12, 20, 26, 36]; their total 94 divided by 4 gives 23.5.

Method / explanation: The adjusted values are [12, 20, 26, 36]; their total 94 divided by 4 gives 23.5.

Trap: Apply both changes to the total before dividing.

Q29. Answer: A | Algebra

If $3x + 4 = 25$, what is x ?

A. 7 - CORRECT: Correct. Subtract 4 and divide by 3: $x = 7$.

B. 6 - INCORRECT: Incorrect: 6 does not follow the stated data. Subtract 4 and divide by 3: $x = 7$.

C. 8 - INCORRECT: Incorrect: 8 is an intermediate, sign or operation error. Subtract 4 and divide by 3: $x = 7$.

D. 21 - INCORRECT: Incorrect: 21 ignores a condition or uses the wrong base. Subtract 4 and divide by 3: $x = 7$.

Method / explanation: Subtract 4 and divide by 3: $x = 7$.

Trap: Undo addition before division.

Q30. Answer: B | Algebra

If $2(x - 2) = 8$, what is x when x is a positive integer and the displayed equality is supplemented by $x + 2 = 8$?

A. 5 - INCORRECT: Incorrect: 5 does not follow the stated data. The second equation fixes x : $x + 2 = 8$, hence $x = 6$.

B. 6 - CORRECT: Correct. The second equation fixes x : $x + 2 = 8$, hence $x = 6$.

C. 7 - INCORRECT: Incorrect: 7 is an intermediate, sign or operation error. The second equation fixes x : $x + 2 = 8$, hence $x = 6$.

D. 12 - INCORRECT: Incorrect: 12 ignores a condition or uses the wrong base. The second equation fixes x : $x + 2 = 8$, hence $x = 6$.

Method / explanation: The second equation fixes x : $x + 2 = 8$, hence $x = 6$.

Trap: An identity alone cannot determine x ; use the independent equation.

Q31. Answer: C | Algebra

The sum of two consecutive integers is 21. What is the smaller integer?

A. 11 - INCORRECT: Incorrect: 11 does not follow the stated data. Let the integers be x and $x+1$. Then $2x+1=21$, so $x=10$.

B. 9 - INCORRECT: Incorrect: 9 is an intermediate, sign or operation error. Let the integers be x and $x+1$. Then $2x+1=21$, so $x=10$.

C. 10 - CORRECT: Correct. Let the integers be x and $x+1$. Then $2x+1=21$, so $x=10$.

D. 21 - INCORRECT: Incorrect: 21 ignores a condition or uses the wrong base. Let the integers be x and $x+1$. Then $2x+1=21$, so $x=10$.

Method / explanation: Let the integers be x and $x+1$. Then $2x+1=21$, so $x=10$.

Trap: The question asks for the smaller integer.

Q32. Answer: D | Algebra

Five years ago A was half of A's age 14 years from now. What is A's present age?

A. 19 - INCORRECT: Incorrect: 19 does not follow the stated data. Let present age be x . Then $x-5 = (x+14)/2$; solving gives $x=24$.

B. 29 - INCORRECT: Incorrect: 29 is an intermediate, sign or operation error. Let present age be x . Then $x-5 = (x+14)/2$; solving gives $x=24$.

C. 12 - INCORRECT: Incorrect: 12 ignores a condition or uses the wrong base. Let present age be x . Then $x-5 = (x+14)/2$; solving gives $x=24$.

D. 24 - CORRECT: Correct. Let present age be x . Then $x-5 = (x+14)/2$; solving gives $x=24$.

Method / explanation: Let present age be x . Then $x-5 = (x+14)/2$; solving gives $x=24$.

Trap: Translate both time references from the present.

Q33. Answer: A | Algebra

What is the positive solution of $x^2 = 64$?

A. 8 - CORRECT: Correct. $x = \text{plus or minus } 8$; the positive solution is 8.

- B. -8 - INCORRECT: Incorrect: -8 does not follow the stated data. $x = \text{plus or minus } 8$; the positive solution is 8.
- C. 64 - INCORRECT: Incorrect: 64 is an intermediate, sign or operation error. $x = \text{plus or minus } 8$; the positive solution is 8.
- D. 9 - INCORRECT: Incorrect: 9 ignores a condition or uses the wrong base. $x = \text{plus or minus } 8$; the positive solution is 8.

Method / explanation: $x = \text{plus or minus } 8$; the positive solution is 8.

Trap: The word positive removes the negative root.

Q34. Answer: B | Algebra

Which value is the least integer satisfying $2x + 1 > 18$?

- A. 8 - INCORRECT: Incorrect: 8 does not follow the stated data. $2x > 17$, so $x > 8.5$; the least integer is 9.
- B. 9 - CORRECT: Correct. $2x > 17$, so $x > 8.5$; the least integer is 9.
- C. 10 - INCORRECT: Incorrect: 10 is an intermediate, sign or operation error. $2x > 17$, so $x > 8.5$; the least integer is 9.
- D. 18 - INCORRECT: Incorrect: 18 ignores a condition or uses the wrong base. $2x > 17$, so $x > 8.5$; the least integer is 9.

Method / explanation: $2x > 17$, so $x > 8.5$; the least integer is 9.

Trap: A strict inequality excludes the boundary.

Q35. Answer: C | Percentages

A grant equals 20% of Rs 320. An administrative surcharge of 7% is then applied to the grant. What is the final amount?

- A. 64 - INCORRECT: Incorrect: 64 does not follow the stated data. First grant = $20/100 \times 320 = 64$; then multiply by 107/100 to get 68.48.
- B. 61.480000000000004 - INCORRECT: Incorrect: 61.480000000000004 is an intermediate, sign or operation error. First grant = $20/100 \times 320 = 64$; then multiply by 107/100 to get 68.48.
- C. 68.48 - CORRECT: Correct. First grant = $20/100 \times 320 = 64$; then multiply by 107/100 to get 68.48.
- D. 86.4 - INCORRECT: Incorrect: 86.4 ignores a condition or uses the wrong base. First grant = $20/100 \times 320 = 64$; then multiply by 107/100 to get 68.48.

Method / explanation: First grant = $20/100 \times 320 = 64$; then multiply by 107/100 to get 68.48.

Trap: The surcharge applies to the grant, not to the original amount.

Q36. Answer: D | Percentages

A value of 1000 rises by 10% and then falls by 5%. What is the final value?

- A. 1050 - INCORRECT: Incorrect: 1050 does not follow the stated data. Successive multipliers give $1000 \times 1.10 \times 0.95 = 1045.0$.
- B. 1050.0 - INCORRECT: Incorrect: 1050.0 is an intermediate, sign or operation error. Successive multipliers give $1000 \times 1.10 \times 0.95 = 1045.0$.
- C. 1055.0 - INCORRECT: Incorrect: 1055.0 ignores a condition or uses the wrong base. Successive multipliers give $1000 \times 1.10 \times 0.95 = 1045.0$.
- D. 1045.0 - CORRECT: Correct. Successive multipliers give $1000 \times 1.10 \times 0.95 = 1045.0$.

Method / explanation: Successive multipliers give $1000 \times 1.10 \times 0.95 = 1045.0$.

Trap: Successive percentages are not simply netted.

Q37. Answer: A | Ratios

Rs 540 is divided in the ratio 4:3. What is the first share?

- A. 308.57142857142856 - CORRECT: Correct. First share = $540 \times 4/(4+3) = 308.57142857142856$.
- B. 304.57142857142856 - INCORRECT: Incorrect: 304.57142857142856 does not follow the stated data. First share = $540 \times 4/(4+3) = 308.57142857142856$.
- C. 312.57142857142856 - INCORRECT: Incorrect: 312.57142857142856 is an intermediate, sign or operation error. First share = $540 \times 4/(4+3) = 308.57142857142856$.
- D. 77.14285714285714 - INCORRECT: Incorrect: 77.14285714285714 ignores a condition or uses the wrong base. First share = $540 \times 4/(4+3) = 308.57142857142856$.

Method / explanation: First share = $540 \times 4/(4+3) = 308.57142857142856$.

Trap: Use ratio parts, not either raw ratio number.

Q38. Answer: B | Averages

Groups of 22 and 32 have averages 42 and 62. What is their combined average?

A. 52.0 - INCORRECT: Incorrect: 52.0 does not follow the stated data. Weighted mean = $(22 \times 42 + 32 \times 62) / (54) = 53.85$.

B. 53.85 - CORRECT: Correct. Weighted mean = $(22 \times 42 + 32 \times 62) / (54) = 53.85$.

C. 52.85 - INCORRECT: Incorrect: 52.85 is an intermediate, sign or operation error. Weighted mean = $(22 \times 42 + 32 \times 62) / (54) = 53.85$.

D. 54.85 - INCORRECT: Incorrect: 54.85 ignores a condition or uses the wrong base. Weighted mean = $(22 \times 42 + 32 \times 62) / (54) = 53.85$.

Method / explanation: Weighted mean = $(22 \times 42 + 32 \times 62) / (54) = 53.85$.

Trap: Do not average group averages without weights.

Q39. Answer: C | Percentages

An article costs Rs 600, is marked for a 17% profit, and then receives a 6% rebate on that marked selling price. What is the realised price?

A. 702.0 - INCORRECT: Incorrect: 702.0 does not follow the stated data. Marked selling price=702.0; realised price= $702.0 \times 94 / 100 = 659.88$.

B. 666.0 - INCORRECT: Incorrect: 666.0 is an intermediate, sign or operation error. Marked selling price=702.0; realised price= $702.0 \times 94 / 100 = 659.88$.

C. 659.88 - CORRECT: Correct. Marked selling price=702.0; realised price= $702.0 \times 94 / 100 = 659.88$.

D. 665.88 - INCORRECT: Incorrect: 665.88 ignores a condition or uses the wrong base. Marked selling price=702.0; realised price= $702.0 \times 94 / 100 = 659.88$.

Method / explanation: Marked selling price=702.0; realised price= $702.0 \times 94 / 100 = 659.88$.

Trap: Profit and rebate use different successive bases.

Q40. Answer: D | Percentages

A marked price of Rs 1000 is discounted by 12%. What is the sale price?

A. 988 - INCORRECT: Incorrect: 988 does not follow the stated data. Sale price = $1000 \times 88 / 100 = 880.0$.

B. 1120.0 - INCORRECT: Incorrect: 1120.0 is an intermediate, sign or operation error. Sale price = $1000 \times 88 / 100 = 880.0$.

C. 868.0 - INCORRECT: Incorrect: 868.0 ignores a condition or uses the wrong base. Sale price = $1000 \times 88 / 100 = 880.0$.

D. 880.0 - CORRECT: Correct. Sale price = $1000 \times 88 / 100 = 880.0$.

Method / explanation: Sale price = $1000 \times 88 / 100 = 880.0$.

Trap: A percentage discount is not a rupee subtraction.

Q41. Answer: A | Ratios

A mixture contains milk and water in the ratio 5:5. In 40 litres, how many litres are milk?

A. 20.0 - CORRECT: Correct. Milk = $40 \times 5 / (5+5) = 20.0$.

B. 19.0 - INCORRECT: Incorrect: 19.0 does not follow the stated data. Milk = $40 \times 5 / (5+5) = 20.0$.

C. 4.0 - INCORRECT: Incorrect: 4.0 is an intermediate, sign or operation error. Milk = $40 \times 5 / (5+5) = 20.0$.

D. 21.0 - INCORRECT: Incorrect: 21.0 ignores a condition or uses the wrong base. Milk = $40 \times 5 / (5+5) = 20.0$.

Method / explanation: Milk = $40 \times 5 / (5+5) = 20.0$.

Trap: The first term of the ratio denotes milk.

Q42. Answer: B | Averages

The average of 7 values is 42. After adding 54, what is the new average?

A. 42 - INCORRECT: Incorrect: 42 does not follow the stated data. Old total 294; new average = $(294+54) / 8 = 43.50$.

B. 43.5 - CORRECT: Correct. Old total 294; new average = $(294+54) / 8 = 43.50$.

C. 48.0 - INCORRECT: Incorrect: 48.0 is an intermediate, sign or operation error. Old total 294; new average = $(294+54)/8 = 43.50$.
D. 44.5 - INCORRECT: Incorrect: 44.5 ignores a condition or uses the wrong base. Old total 294; new average = $(294+54)/8 = 43.50$.
Method / explanation: Old total 294; new average = $(294+54)/8 = 43.50$.
Trap: Recover the old total before adding a value.

Q43. Answer: C | Time and Work

A can finish a task in 10 days and B in 14 days. How many days will they take together?

- A. 24 - INCORRECT: Incorrect: 24 does not follow the stated data. Combined rate = $1/10+1/14$; time = $140/(24) = 5.83$ days.
B. 12.0 - INCORRECT: Incorrect: 12.0 is an intermediate, sign or operation error. Combined rate = $1/10+1/14$; time = $140/(24) = 5.83$ days.
C. 5.83 - CORRECT: Correct. Combined rate = $1/10+1/14$; time = $140/(24) = 5.83$ days.
D. 4 - INCORRECT: Incorrect: 4 ignores a condition or uses the wrong base. Combined rate = $1/10+1/14$; time = $140/(24) = 5.83$ days.

Method / explanation: Combined rate = $1/10+1/14$; time = $140/(24) = 5.83$ days.

Trap: Add rates, not times.

Q44. Answer: D | Time and Work

14 workers finish a job in 15 days at equal efficiency. How many days will 20 workers take?

- A. 15 - INCORRECT: Incorrect: 15 does not follow the stated data. Worker-days stay constant: $14 \times 15 = 20 \times d$, so $d = 10.50$.
B. 21.43 - INCORRECT: Incorrect: 21.43 is an intermediate, sign or operation error. Worker-days stay constant: $14 \times 15 = 20 \times d$, so $d = 10.50$.
C. 11.5 - INCORRECT: Incorrect: 11.5 ignores a condition or uses the wrong base. Worker-days stay constant: $14 \times 15 = 20 \times d$, so $d = 10.50$.
D. 10.5 - CORRECT: Correct. Worker-days stay constant: $14 \times 15 = 20 \times d$, so $d = 10.50$.

Method / explanation: Worker-days stay constant: $14 \times 15 = 20 \times d$, so $d = 10.50$.

Trap: For fixed work, workers and days vary inversely.

Q45. Answer: A | Time and Work

Two pipes fill a tank in 8 and 10 hours. If opened together, how long do they take?

- A. 4.44 - CORRECT: Correct. Rates add: $1/8+1/10$; reciprocal gives 4.44 hours.
B. 18 - INCORRECT: Incorrect: 18 does not follow the stated data. Rates add: $1/8+1/10$; reciprocal gives 4.44 hours.
C. 9.0 - INCORRECT: Incorrect: 9.0 is an intermediate, sign or operation error. Rates add: $1/8+1/10$; reciprocal gives 4.44 hours.
D. 5.44 - INCORRECT: Incorrect: 5.44 ignores a condition or uses the wrong base. Rates add: $1/8+1/10$; reciprocal gives 4.44 hours.

Method / explanation: Rates add: $1/8+1/10$; reciprocal gives 4.44 hours.

Trap: Both are inlet pipes.

Q46. Answer: B | Time and Work

An inlet fills a tank in 7 hours while a leak empties it in 22 hours. How long to fill together?

- A. 5.31 - INCORRECT: Incorrect: 5.31 does not follow the stated data. Net rate = $1/7-1/22$; time = 10.27 hours.
B. 10.27 - CORRECT: Correct. Net rate = $1/7-1/22$; time = 10.27 hours.
C. 15 - INCORRECT: Incorrect: 15 is an intermediate, sign or operation error. Net rate = $1/7-1/22$; time = 10.27 hours.
D. 7 - INCORRECT: Incorrect: 7 ignores a condition or uses the wrong base. Net rate = $1/7-1/22$; time = 10.27 hours.

Method / explanation: Net rate = $1/7-1/22$; time = 10.27 hours.

Trap: An outlet rate must be subtracted.

Q47. Answer: C | Time and Work

A is 4 times as efficient as B. If B alone takes 40 days, how many days does A take?

- A. 40 - INCORRECT: Incorrect: 40 does not follow the stated data. Time is inverse to efficiency: $40/4=10$ days.
- B. 4 - INCORRECT: Incorrect: 4 is an intermediate, sign or operation error. Time is inverse to efficiency: $40/4=10$ days.
- C. 10 - CORRECT: Correct. Time is inverse to efficiency: $40/4=10$ days.
- D. 14 - INCORRECT: Incorrect: 14 ignores a condition or uses the wrong base. Time is inverse to efficiency: $40/4=10$ days.
- Method / explanation: Time is inverse to efficiency: $40/4=10$ days.
- Trap: Greater efficiency means less time.

Q48. Answer: D | Time and Work

A team completes 0.40 of a job. What fraction remains, rounded to two decimals?

- A. 0.4 - INCORRECT: Incorrect: 0.4 does not follow the stated data. Remaining work = $1 - 0.40 = 0.60$.
- B. 1.4 - INCORRECT: Incorrect: 1.4 is an intermediate, sign or operation error. Remaining work = $1 - 0.40 = 0.60$.
- C. 0.7 - INCORRECT: Incorrect: 0.7 ignores a condition or uses the wrong base. Remaining work = $1 - 0.40 = 0.60$.
- D. 0.6 - CORRECT: Correct. Remaining work = $1 - 0.40 = 0.60$.

Method / explanation: Remaining work = $1 - 0.40 = 0.60$.

Trap: Completed and remaining fractions sum to one.

Q49. Answer: A | Speed and Distance

A vehicle's total journey time is 5 hours, including a halt of 30 minutes. It moves at 55 km/h whenever in motion. What distance is covered?

- A. 247.5 - CORRECT: Correct. Moving time = $5 - 30/60 = 4.5$ hours; distance = $55 \times 4.5 = 247.5$ km.
- B. 275 - INCORRECT: Incorrect: 275 does not follow the stated data. Moving time = $5 - 30/60 = 4.5$ hours; distance = $55 \times 4.5 = 247.5$ km.
- C. 192.5 - INCORRECT: Incorrect: 192.5 is an intermediate, sign or operation error. Moving time = $5 - 30/60 = 4.5$ hours; distance = $55 \times 4.5 = 247.5$ km.
- D. 302.5 - INCORRECT: Incorrect: 302.5 ignores a condition or uses the wrong base. Moving time = $5 - 30/60 = 4.5$ hours; distance = $55 \times 4.5 = 247.5$ km.

Method / explanation: Moving time = $5 - 30/60 = 4.5$ hours; distance = $55 \times 4.5 = 247.5$ km.

Trap: Remove halt time before multiplying by speed.

Q50. Answer: B | Speed and Distance

Over 280 km, how many hours are saved by increasing speed from 42 to 62 km/h?

- A. 14.0 - INCORRECT: Incorrect: 14.0 does not follow the stated data. Time saved = $280/42 - 280/62 = 2.15$ hours.
- B. 2.15 - CORRECT: Correct. Time saved = $280/42 - 280/62 = 2.15$ hours.
- C. 3.15 - INCORRECT: Incorrect: 3.15 is an intermediate, sign or operation error. Time saved = $280/42 - 280/62 = 2.15$ hours.
- D. 1.15 - INCORRECT: Incorrect: 1.15 ignores a condition or uses the wrong base. Time saved = $280/42 - 280/62 = 2.15$ hours.

Method / explanation: Time saved = $280/42 - 280/62 = 2.15$ hours.

Trap: Subtract travel times, not speeds.

Q51. Answer: C | Speed and Distance

A 200-metre train moves at 54 km/h. How many seconds does it take to clear a 100-metre platform?

- A. 13.33 - INCORRECT: Incorrect: 13.33 does not follow the stated data. Total distance = $200 + 100$; $54 \text{ km/h} = 15 \text{ m/s}$; time = 20.00 s.
- B. 22.0 - INCORRECT: Incorrect: 22.0 is an intermediate, sign or operation error. Total distance = $200 + 100$; $54 \text{ km/h} = 15 \text{ m/s}$; time = 20.00 s.
- C. 20.0 - CORRECT: Correct. Total distance = $200 + 100$; $54 \text{ km/h} = 15 \text{ m/s}$; time = 20.00 s.
- D. 18.0 - INCORRECT: Incorrect: 18.0 ignores a condition or uses the wrong base. Total distance = $200 + 100$; $54 \text{ km/h} = 15 \text{ m/s}$; time = 20.00 s.

Method / explanation: Total distance = $200 + 100$; $54 \text{ km/h} = 15 \text{ m/s}$; time = 20.00 s.

Trap: To clear a platform, use train length plus platform length.

Q52. Answer: D | Speed and Distance

A boat's still-water speed is 12 km/h and stream speed is 4 km/h. How long does it take to travel 16 km upstream?

A. 1.0 - INCORRECT: Incorrect: 1.0 does not follow the stated data. Upstream speed= $12-4=8$ km/h; time= $16/8=2$ hours.

B. 8 - INCORRECT: Incorrect: 8 is an intermediate, sign or operation error. Upstream speed= $12-4=8$ km/h; time= $16/8=2$ hours.

C. 1.3333333333333333 - INCORRECT: Incorrect: 1.3333333333333333 ignores a condition or uses the wrong base. Upstream speed= $12-4=8$ km/h; time= $16/8=2$ hours.

D. 2 - CORRECT: Correct. Upstream speed= $12-4=8$ km/h; time= $16/8=2$ hours.

Method / explanation: Upstream speed= $12-4=8$ km/h; time= $16/8=2$ hours.

Trap: First obtain upstream speed, then divide distance by that speed.

Q53. Answer: A | Speed and Distance

A runner completes 5 laps of a square field of side 22 m in 10 minutes. What is the runner's speed in metres per minute?

A. 44.0 - CORRECT: Correct. Total distance= $5 \times 4 \times 22$; divide by 10 minutes to get 44.0 m/min.

B. 88 - INCORRECT: Incorrect: 88 does not follow the stated data. Total distance= $5 \times 4 \times 22$; divide by 10 minutes to get 44.0 m/min.

C. 48.4 - INCORRECT: Incorrect: 48.4 is an intermediate, sign or operation error. Total distance= $5 \times 4 \times 22$; divide by 10 minutes to get 44.0 m/min.

D. 17.6 - INCORRECT: Incorrect: 17.6 ignores a condition or uses the wrong base. Total distance= $5 \times 4 \times 22$; divide by 10 minutes to get 44.0 m/min.

Method / explanation: Total distance= $5 \times 4 \times 22$; divide by 10 minutes to get 44.0 m/min.

Trap: Compute total distance and total time before cancelling common factors.

Q54. Answer: B | Speed and Distance

A car covers 120 km at 42 km/h and 180 km at 62 km/h. What is average speed?

A. 52.0 - INCORRECT: Incorrect: 52.0 does not follow the stated data. Average speed = total distance/total time = $300/(120/42+180/62) = 52.08$.

B. 52.08 - CORRECT: Correct. Average speed = total distance/total time = $300/(120/42+180/62) = 52.08$.

C. 54.08 - INCORRECT: Incorrect: 54.08 is an intermediate, sign or operation error. Average speed = total distance/total time = $300/(120/42+180/62) = 52.08$.

D. 50.08 - INCORRECT: Incorrect: 50.08 ignores a condition or uses the wrong base. Average speed = total distance/total time = $300/(120/42+180/62) = 52.08$.

Method / explanation: Average speed = total distance/total time = $300/(120/42+180/62) = 52.08$.

Trap: Speeds need time weights.

Q55. Answer: C | Data Interpretation

A table reports district outputs (A=90, B=103, C=116, D=129). What is the total?

A. 428 - INCORRECT: Incorrect: 428 does not follow the stated data. Add all four entries: $90+103+116+129=438$.

B. 448 - INCORRECT: Incorrect: 448 is an intermediate, sign or operation error. Add all four entries: $90+103+116+129=438$.

C. 438 - CORRECT: Correct. Add all four entries: $90+103+116+129=438$.

D. 129 - INCORRECT: Incorrect: 129 ignores a condition or uses the wrong base. Add all four entries: $90+103+116+129=438$.

Method / explanation: Add all four entries: $90+103+116+129=438$.

Trap: Use every row exactly once.

Q56. Answer: D | Data Interpretation

Using the table (A=90, B=103, C=116, D=129), what is the difference between the highest and lowest values?

A. 129 - INCORRECT: Incorrect: 129 does not follow the stated data. Range = $129-90=39$.

- B. 90 - INCORRECT: Incorrect: 90 is an intermediate, sign or operation error. Range = $129-90=39$.
- C. 219 - INCORRECT: Incorrect: 219 ignores a condition or uses the wrong base. Range = $129-90=39$.
- D. 39 - CORRECT: Correct. Range = $129-90=39$.

Method / explanation: Range = $129-90=39$.

Trap: Range is a difference, not an endpoint.

Q57. Answer: A | Data Interpretation

Using the table (A=90, B=103, C=116, D=129), what is the average of the four values?

- A. 109.5 - CORRECT: Correct. Average = $438/4=109.5$.
- B. 438 - INCORRECT: Incorrect: 438 does not follow the stated data. Average = $438/4=109.5$.
- C. 219.0 - INCORRECT: Incorrect: 219.0 is an intermediate, sign or operation error. Average = $438/4=109.5$.
- D. 39 - INCORRECT: Incorrect: 39 ignores a condition or uses the wrong base. Average = $438/4=109.5$.

Method / explanation: Average = $438/4=109.5$.

Trap: Divide the total by four.

Q58. Answer: B | Data Interpretation

Using the table (A=90, B=103, C=116, D=129), by what percentage does D exceed A?

- A. 30.23 - INCORRECT: Incorrect: 30.23 does not follow the stated data. Percentage increase = $(129-90)/90 \times 100$.
- B. 43.33 - CORRECT: Correct. Percentage increase = $(129-90)/90 \times 100$.
- C. 39 - INCORRECT: Incorrect: 39 is an intermediate, sign or operation error. Percentage increase = $(129-90)/90 \times 100$.
- D. 143.33 - INCORRECT: Incorrect: 143.33 ignores a condition or uses the wrong base. Percentage increase = $(129-90)/90 \times 100$.

Method / explanation: Percentage increase = $(129-90)/90 \times 100$.

Trap: Use the original value A as denominator.

Q59. Answer: C | Data Interpretation

Year 1 values are A=90, B=103, C=116, D=129; Year 2 values are A=112, B=120, C=128, D=136. Which district has the largest absolute increase?

- A. B - INCORRECT: Incorrect: B does not follow the stated data. The increases are 22, 17, 12 and 7; A is largest.
- B. C - INCORRECT: Incorrect: C is an intermediate, sign or operation error. The increases are 22, 17, 12 and 7; A is largest.
- C. A - CORRECT: Correct. The increases are 22, 17, 12 and 7; A is largest.
- D. D - INCORRECT: Incorrect: D ignores a condition or uses the wrong base. The increases are 22, 17, 12 and 7; A is largest.

Method / explanation: The increases are 22, 17, 12 and 7; A is largest.

Trap: Check absolute increases before comparing totals.

Q60. Answer: D | Data Interpretation

Using Year 2 (A=112, B=120, C=128, D=136), what fraction of the total belongs to B, rounded to two decimals?

- A. 0.27 - INCORRECT: Incorrect: 0.27 does not follow the stated data. B's share = $120/496 = 0.24$.
- B. 4.13 - INCORRECT: Incorrect: 4.13 is an intermediate, sign or operation error. B's share = $120/496 = 0.24$.
- C. 0.23 - INCORRECT: Incorrect: 0.23 ignores a condition or uses the wrong base. B's share = $120/496 = 0.24$.
- D. 0.24 - CORRECT: Correct. B's share = $120/496 = 0.24$.

Method / explanation: B's share = $120/496 = 0.24$.

Trap: The denominator is the same-year total.

Q61. Answer: A | Data Interpretation

Using Year 1 (A=90, B=103, C=116, D=129), if each unit earns Rs 4, what is D's revenue?

- A. 516 - CORRECT: Correct. Revenue = D units x rate = 129 x 4.
- B. 133 - INCORRECT: Incorrect: 133 does not follow the stated data. Revenue = D units x rate = 129 x 4.
- C. 129 - INCORRECT: Incorrect: 129 is an intermediate, sign or operation error. Revenue = D units x rate = 129 x 4.
- D. 1752 - INCORRECT: Incorrect: 1752 ignores a condition or uses the wrong base. Revenue = D units x rate = 129 x 4.
- Method / explanation: Revenue = D units x rate = 129 x 4.
- Trap: Use D's value, not total output.

Q62. Answer: B | Data Interpretation

Using Year 1 (A=90, B=103, C=116, D=129), what is the ratio A:D in simplest form?

- A. 129:90 - INCORRECT: Incorrect: 129:90 does not follow the stated data. Divide 90:129 by gcd 3.
- B. 30:43 - CORRECT: Correct. Divide 90:129 by gcd 3.
- C. 180:258 - INCORRECT: Incorrect: 180:258 is an intermediate, sign or operation error. Divide 90:129 by gcd 3.
- D. 219:1 - INCORRECT: Incorrect: 219:1 ignores a condition or uses the wrong base. Divide 90:129 by gcd 3.

Method / explanation: Divide 90:129 by gcd 3.

Trap: Preserve the requested A-to-D order.

Q63. Answer: C | Logical Reasoning

All surgeons are licensed. No person lacking that status may operate. Which conclusion necessarily follows?

- A. Every licensed person is one of the surgeons. - INCORRECT: Incorrect: Every licensed person is one of the surgeons. does not follow the stated data. All surgeons possess the stated status, so none can simultaneously lack it.
- B. Some surgeons lack that status. - INCORRECT: Incorrect: Some surgeons lack that status. is an intermediate, sign or operation error. All surgeons possess the stated status, so none can simultaneously lack it.
- C. Every surgeon is outside the class lacking that status. - CORRECT: Correct. All surgeons possess the stated status, so none can simultaneously lack it.
- D. No licensed person may operate. - INCORRECT: Incorrect: No licensed person may operate. ignores a condition or uses the wrong base. All surgeons possess the stated status, so none can simultaneously lack it.

Method / explanation: All surgeons possess the stated status, so none can simultaneously lack it.

Trap: Do not reverse a universal proposition.

Q64. Answer: D | Logical Reasoning

If each letter is shifted forward by 2, how is INDIA coded?

- A. INDIA - INCORRECT: Incorrect: INDIA does not follow the stated data. Shift each letter independently by 2: INDIA -> KPFKC.
- B. CKFPK - INCORRECT: Incorrect: CKFPK is an intermediate, sign or operation error. Shift each letter independently by 2: INDIA -> KPFKC.
- C. JODJB - INCORRECT: Incorrect: JODJB ignores a condition or uses the wrong base. Shift each letter independently by 2: INDIA -> KPFKC.
- D. KPFKC - CORRECT: Correct. Shift each letter independently by 2: INDIA -> KPFKC.

Method / explanation: Shift each letter independently by 2: INDIA -> KPFKC.

Trap: Do not reverse the word.

Q65. Answer: A | Logical Reasoning

Complete the series: 4, 6, 10, 16, ?

- A. 24 - CORRECT: Correct. Successive increments are 2, 4, 6 and 8; next term is 24.
- B. 23 - INCORRECT: Incorrect: 23 does not follow the stated data. Successive increments are 2, 4, 6 and 8; next term is 24.
- C. 25 - INCORRECT: Incorrect: 25 is an intermediate, sign or operation error. Successive increments are 2, 4, 6 and 8; next term is 24.

D. 22 - INCORRECT: Incorrect: 22 ignores a condition or uses the wrong base. Successive increments are 2, 4, 6 and 8; next term is 24.

Method / explanation: Successive increments are 2, 4, 6 and 8; next term is 24.

Trap: Track differences, not ratios.

Q66. Answer: B | Logical Reasoning

Rohan is the child of Leela's only son. How is Rohan related to Leela?

A. Child - INCORRECT: Incorrect: Child does not follow the stated data. The only son is the parent, so Rohan is the elder's grandchild.

B. Grandchild - CORRECT: Correct. The only son is the parent, so Rohan is the elder's grandchild.

C. Niece or nephew - INCORRECT: Incorrect: Niece or nephew is an intermediate, sign or operation error. The only son is the parent, so Rohan is the elder's grandchild.

D. Sibling - INCORRECT: Incorrect: Sibling ignores a condition or uses the wrong base. The only son is the parent, so Rohan is the elder's grandchild.

Method / explanation: The only son is the parent, so Rohan is the elder's grandchild.

Trap: Resolve one relationship at a time.

Q67. Answer: C | Logical Reasoning

Some villages are coastal settlements. All coastal settlements are mapped. Which conclusion follows?

A. All villages are mapped - INCORRECT: Incorrect: All villages are mapped does not follow the stated data. The members of villages that are coastal settlements inherit the property mapped.

B. No villages are mapped - INCORRECT: Incorrect: No villages are mapped is an intermediate, sign or operation error. The members of villages that are coastal settlements inherit the property mapped.

C. Some villages are mapped - CORRECT: Correct. The members of villages that are coastal settlements inherit the property mapped.

D. Some mapped are not coastal settlements - INCORRECT: Incorrect: Some mapped are not coastal settlements ignores a condition or uses the wrong base. The members of villages that are coastal settlements inherit the property mapped.

Method / explanation: The members of villages that are coastal settlements inherit the property mapped.

Trap: Do not universalise a particular premise.

Q68. Answer: D | Logical Reasoning

In a code, GOVERN is written as HPWFSO by shifting each letter forward once. How is RIVER written?

A. RIVER - INCORRECT: Incorrect: RIVER does not follow the stated data. Shift every letter of RIVER forward once to obtain SJWFS.

B. SFWJS - INCORRECT: Incorrect: SFWJS is an intermediate, sign or operation error. Shift every letter of RIVER forward once to obtain SJWFS.

C. AAAAA - INCORRECT: Incorrect: AAAAA ignores a condition or uses the wrong base. Shift every letter of RIVER forward once to obtain SJWFS.

D. SJWFS - CORRECT: Correct. Shift every letter of RIVER forward once to obtain SJWFS.

Method / explanation: Shift every letter of RIVER forward once to obtain SJWFS.

Trap: Apply the same shift to every letter.

Q69. Answer: A | Arrangements

P, Q, R and S sit in a row. P is left of Q; R is right of Q; S is left of P. Who is second from the left?

A. P - CORRECT: Correct. The only order consistent with all relations is S-P-Q-R.

B. Q - INCORRECT: Incorrect: Q does not follow the stated data. The only order consistent with all relations is S-P-Q-R.

C. R - INCORRECT: Incorrect: R is an intermediate, sign or operation error. The only order consistent with all relations is S-P-Q-R.

D. S - INCORRECT: Incorrect: S ignores a condition or uses the wrong base. The only order consistent with all relations is S-P-Q-R.

Method / explanation: The only order consistent with all relations is S-P-Q-R.

Trap: Build the full order before selecting a position.

Q70. Answer: B | Arrangements

Four tasks Audit, Budget, Consultation and Draft occur consecutively. Audit precedes Budget; Consultation follows Budget; Draft precedes Audit. Which task is third?

A. Audit - INCORRECT: Incorrect: Audit does not follow the stated data. The forced order is Draft-Audit-Budget-Consultation, so Budget is third.

B. Budget - CORRECT: Correct. The forced order is Draft-Audit-Budget-Consultation, so Budget is third.

C. Consultation - INCORRECT: Incorrect: Consultation is an intermediate, sign or operation error. The forced order is Draft-Audit-Budget-Consultation, so Budget is third.

D. Draft - INCORRECT: Incorrect: Draft ignores a condition or uses the wrong base. The forced order is Draft-Audit-Budget-Consultation, so Budget is third.

Method / explanation: The forced order is Draft-Audit-Budget-Consultation, so Budget is third.

Trap: Precedes means earlier, not immediately earlier unless stated.

Q71. Answer: C | Arrangements

Five persons stand by height. Asha is taller than Bina but shorter than Charu. Diya is shorter than Bina. Esha is taller than Charu. Who is tallest?

A. Charu - INCORRECT: Incorrect: Charu does not follow the stated data. The chain is Esha > Charu > Asha > Bina > Diya.

B. Asha - INCORRECT: Incorrect: Asha is an intermediate, sign or operation error. The chain is Esha > Charu > Asha > Bina > Diya.

C. Esha - CORRECT: Correct. The chain is Esha > Charu > Asha > Bina > Diya.

D. Diya - INCORRECT: Incorrect: Diya ignores a condition or uses the wrong base. The chain is Esha > Charu > Asha > Bina > Diya.

Method / explanation: The chain is Esha > Charu > Asha > Bina > Diya.

Trap: Combine all comparisons into one chain.

Q72. Answer: D | Arrangements

Hearing is after Drafting but before Order. Survey is before Drafting. Which is earliest?

A. Drafting - INCORRECT: Incorrect: Drafting does not follow the stated data. The order is Survey, Drafting, Hearing, Order.

B. Hearing - INCORRECT: Incorrect: Hearing is an intermediate, sign or operation error. The order is Survey, Drafting, Hearing, Order.

C. Order - INCORRECT: Incorrect: Order ignores a condition or uses the wrong base. The order is Survey, Drafting, Hearing, Order.

D. Survey - CORRECT: Correct. The order is Survey, Drafting, Hearing, Order.

Method / explanation: The order is Survey, Drafting, Hearing, Order.

Trap: Translate every before/after relation consistently.

Q73. Answer: A | Directions

A person faces North and turns right 2 time(s), each by 90 degrees. Which direction is faced?

A. South - CORRECT: Correct. Each right turn advances one step clockwise; 2 turn(s) gives South.

B. North - INCORRECT: Incorrect: North does not follow the stated data. Each right turn advances one step clockwise; 2 turn(s) gives South.

C. East - INCORRECT: Incorrect: East is an intermediate, sign or operation error. Each right turn advances one step clockwise; 2 turn(s) gives South.

D. West - INCORRECT: Incorrect: West ignores a condition or uses the wrong base. Each right turn advances one step clockwise; 2 turn(s) gives South.

Method / explanation: Each right turn advances one step clockwise; 2 turn(s) gives South.

Trap: Reduce turns modulo four.

Q74. Answer: B | Directions

A walker goes 6 km north and 8 km east. How far is the endpoint from the start?

- A. 14 km - INCORRECT: Incorrect: 14 km does not follow the stated data. Displacement = $\sqrt{6^2+8^2}=10$ km.
- B. 10 km - CORRECT: Correct. Displacement = $\sqrt{6^2+8^2}=10$ km.
- C. 2 km - INCORRECT: Incorrect: 2 km is an intermediate, sign or operation error. Displacement = $\sqrt{6^2+8^2}=10$ km.
- D. 48 km - INCORRECT: Incorrect: 48 km ignores a condition or uses the wrong base. Displacement = $\sqrt{6^2+8^2}=10$ km.

Method / explanation: Displacement = $\sqrt{6^2+8^2}=10$ km.

Trap: Distance walked and displacement differ.

Q75. Answer: C | Directions

A person starts facing South, turns left, then turns left again. Which direction is now faced?

- A. East - INCORRECT: Incorrect: East does not follow the stated data. Two left turns reverse the starting direction, giving North.
- B. South - INCORRECT: Incorrect: South is an intermediate, sign or operation error. Two left turns reverse the starting direction, giving North.
- C. North - CORRECT: Correct. Two left turns reverse the starting direction, giving North.
- D. West - INCORRECT: Incorrect: West ignores a condition or uses the wrong base. Two left turns reverse the starting direction, giving North.

Method / explanation: Two left turns reverse the starting direction, giving North.

Trap: Track orientation after each turn.

Q76. Answer: D | Directions

A point B is 6 km south of A, and C is 6 km east of B. In which direction is C from A?

- A. North-East - INCORRECT: Incorrect: North-East does not follow the stated data. Relative to A, C has positive east and negative north coordinates: south-east.
- B. South-West - INCORRECT: Incorrect: South-West is an intermediate, sign or operation error. Relative to A, C has positive east and negative north coordinates: south-east.
- C. North-West - INCORRECT: Incorrect: North-West ignores a condition or uses the wrong base. Relative to A, C has positive east and negative north coordinates: south-east.
- D. South-East - CORRECT: Correct. Relative to A, C has positive east and negative north coordinates: south-east.

Method / explanation: Relative to A, C has positive east and negative north coordinates: south-east.

Trap: Use a simple coordinate sketch.

Q77. Answer: A | Data Sufficiency

What is x? Statement 1: $x+4=12$. Statement 2: $2x=16$.

- A. Each statement alone is sufficient. - CORRECT: Correct. Statement 1 gives $x=8$; statement 2 also gives $x=8$. Each alone is sufficient.
- B. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Statement 1 gives $x=8$; statement 2 also gives $x=8$. Each alone is sufficient.
- C. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Statement 1 gives $x=8$; statement 2 also gives $x=8$. Each alone is sufficient.
- D. Both statements together are sufficient, but neither alone is sufficient. - INCORRECT: Incorrect: Both statements together are sufficient, but neither alone is sufficient. ignores a condition or uses the wrong base. Statement 1 gives $x=8$; statement 2 also gives $x=8$. Each alone is sufficient.

Method / explanation: Statement 1 gives $x=8$; statement 2 also gives $x=8$. Each alone is sufficient.

Trap: Test each statement separately before combining.

Q78. Answer: B | Data Sufficiency

What is the area of a rectangle? Statement 1: its length is 9 cm. Statement 2: its breadth is 6 cm.

A. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Area requires both length and breadth; neither statement alone is enough.

B. Both statements together are sufficient, but neither alone is sufficient. - CORRECT: Correct. Area requires both length and breadth; neither statement alone is enough.

C. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Area requires both length and breadth; neither statement alone is enough.

D. Each statement alone is sufficient. - INCORRECT: Incorrect: Each statement alone is sufficient. ignores a condition or uses the wrong base. Area requires both length and breadth; neither statement alone is enough.

Method / explanation: Area requires both length and breadth; neither statement alone is enough.

Trap: Do not import an unstated shape relation.

Q79. Answer: C | Data Sufficiency

Is integer n even? Statement 1: n is divisible by 12. Statement 2: n is divisible by 6.

A. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Either divisibility condition independently implies n is even.

B. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Either divisibility condition independently implies n is even.

C. Each statement alone is sufficient. - CORRECT: Correct. Either divisibility condition independently implies n is even.

D. Both statements together are sufficient, but neither alone is sufficient. - INCORRECT: Incorrect: Both statements together are sufficient, but neither alone is sufficient. ignores a condition or uses the wrong base. Either divisibility condition independently implies n is even.

Method / explanation: Either divisibility condition independently implies n is even.

Trap: Sufficiency does not require both statements to be equally strong.

Q80. Answer: D | Data Sufficiency

What is y ? Statement 1: y is positive. Statement 2: $y^2=64$.

A. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Statement 2 gives $y=$ plus or minus 8; statement 1 selects $y=8$. Together they are sufficient.

B. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Statement 2 gives $y=$ plus or minus 8; statement 1 selects $y=8$. Together they are sufficient.

C. Each statement alone is sufficient. - INCORRECT: Incorrect: Each statement alone is sufficient. ignores a condition or uses the wrong base. Statement 2 gives $y=$ plus or minus 8; statement 1 selects $y=8$. Together they are sufficient.

D. Both statements together are sufficient, but neither alone is sufficient. - CORRECT: Correct. Statement 2 gives $y=$ plus or minus 8; statement 1 selects $y=8$. Together they are sufficient.

Method / explanation: Statement 2 gives $y=$ plus or minus 8; statement 1 selects $y=8$. Together they are sufficient.

Trap: A square equation has two real roots unless sign is fixed.

END OF DETAILED SOLUTIONS